IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- SpaceX has achieved historic milestones with Falcon 9. By reusing its first-stage boosters they've been able to greatly reduce the cost of a launch. Our startup has been tasked with predicting the cost of a rocket launch so other companies can make more informed bids against SpaceX.
- The SpaceX launch data was collected from The SpaceX REST API and by web scraping.
- During data wrangling we dealt with missing values and performed preliminary exploration to gain a better understanding of the data.
- In Exploratory data analysis (EDA) we built graphs and used SQL queries to gain valuable insights. We identified the dependent and independent variables for our model.
- We developed Interactive maps and dashboards using Folium and Plotly Dash.
- We perform predictive analysis using the following classification models: Logistic Regression, Support Vector Machine, Decision Trees and K-Nearest Neighbors.
- All four models had the same classification accuracy at 0.83; and the same results for the confusion matrix. Although the models can be considered good at predicting, there might be room for improvement.

Introduction

- SpaceX is a private aerospace manufacturer and space transportation company founded in 2002. Its goal is to revolutionize space technology and enable human colonization of other planets. SpaceX has achieved several milestones, including launching and recovering spacecraft, sending astronauts to the International Space Station, and developing liquid-propellant rockets that reach orbit. (SpaceX – Wikipedia)
- SpaceX is the only private company to have successfully returned a spacecraft from low Earth orbit, first accomplishing this in December 2010. The company advertises Falcon 9 rocket launches on its website at a cost of approximately $62 million, whereas other providers can cost upward of $165 million per launch. By reusing the first stage of its rockets—when they are successfully recovered and landed—SpaceX is able to save millions of dollars.
- If we develop a model that can predict if the first stage of a rocket will land successfully then we could determine the cost of future launches. With this information, competitors could make more informed decisions when bidding against SpaceX.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - The SpaceX launch data was collected from The SpaceX REST API and by scraping the Wikipedia page: “List of Falcon 9 and Falcon Heavy launches”.
- Perform data wrangling:
  - Data wrangling included data cleaning (for missing values) and preliminary exploration to gain a better understanding of the data.
- Perform exploratory data analysis (EDA) using visualization and SQL.
- Perform interactive visual analytics using Folium and Plotly Dash.
- Perform predictive analysis using classification models.
  - Four classification models were built to predict if the first stage of the Falcon 9 will land successfully. Trained the data using Grid Search. Identified the model with the best classification accuracy.

Data Collection

- The SpaceX launch data was gathered from two sources:
  - The SpaceX REST API.
    - We used the Python requests library to perform a GET request using a static URL. The data was obtained in JSON format and then converted into a dataframe.
  - The Wikipedia page: “List of Falcon 9 and Falcon Heavy launches”.
    - We used a GET request to web scrape an HTML table containing Falcon 9 launch records. We applied BeautifulSoup to extract the table and parse the data.

Data Collection - SpaceX API

SpaceX REST API

https://api.spacexdata.com/v4/

launches/past

/rockets

/launchpads

/payloads

/cores

Instead of using the API responses directly, the data was requested using a static URL

Obtain data with a GET request:
response = requests.get(static_URL).json

Successfully receive a .json response:
response.status_code = 200

Turn the content into a dataframe:
data=pd.json_normalize(response.json())

Filter dataframe for Falcon 9 launches

Notebook GitHub URL: IBM-Data-Science-Capstone/Collecting the Data API-api.ipynb at main · elgels/IBM-Data-Science-Capstone

Data Collection - Scraping

Wikipedia page:

"List of Falcon 9 and Falcon Heavy launches"

HTML page was requested using a static URL

Obtain data with a GET request:
response = requests.get(static_URL).text

Use the Python BeautifulSoup package to create an object from the HTML response:
soup = BeautifulSoup(response, 'html.parser')

Find the HTML table with Falcon 9 records:
html_tables = soup.find_all('table')

Iterate through the <th> elements of the table and parse the raw data to convert it into a dataframe

Notebook GitHub URL: [IBM-Data-Science-Capstone/Webscraping Records from Wikipedia.ipynb at main · elgels/IBM-Data-Science-Capstone]

Data Wrangling - Process

- In data wrangling, we followed this process:

Identify Missing Values → Count observations → Identify variables type → Perform value counts → Create variable "Class" (success or failure)

  - The landing pad column has 26 nulls (29% of total).
  - The dataframe consists of 90 rows and 17 columns.
  - The dates ranges from June 2010 through November 2020.
  - 7 variables are String type, 7 are Numeric and 3 are Boolean.
  - Created a new column "Class" to represent landing success (value of 1) or failure (value of 0).

Notebook GitHub URL: [IBM-Data-Science-Capstone/Data Wrangling.ipynb at main · elgels/IBM-Data-Science-Capstone]

Data Wrangling - Variables

String	Description
Date	YYYY-MM-DD
Booster Version	Versions of the Falcon 9 boosters such as F9 v1.1 and F9 v1.1 B1012.
Orbit	Such as LEO (low orbit at an altitude of 2,000 km from Earth), VLEO (very low orbit at below 450 km), GTO (high Earth orbit at about 36,000 km).
Launch Site	Cape Canaveral (CCAFS SLC), Kennedy Space Center (KSC LC 39A) and Vandenberg (VAFB SLC 4E).
Outcome	Codes for successful and failed landings (in ocean, ground and drone ship).
Landing Pad	5 Ids of Landing Pads.
Serial	Serial Numbers such as B1049, B1051 and B1060.

Numeric	Description
Flight Number	Id numbers from 1 through 90.
Payload Mass	Weight (Kg.) of cargo or equipment that the rocket is designed to carry into space.
Flights	Values from 1 through 6.
Block	Incremental improvements made to previous versions. From 1.0 through 5.0.
Reused Count	Values from 0 through 5.
Longitude	Longitude of the launch site.
Latitude	Latitude of the launch site.
Class	0 (if launch failed), 1 (if launch succeeded).

Boolean	Description
Grid Fins	True or False.
Reused	True or False.
Legs	True or False.

EDA and Data Visualization

- We created the following charts:

Scatter Plot	Bar Chart	Line Chart
Flight Number vs Launch Site	Success Rate vs. Orbit Type	Yearly Average Success Rate
Payload vs. Launch Site		
Flight Number vs. Orbit Type		

- Launch sites exhibited different success rates depending on the payload mass they carried, with heavier and lighter missions performing differently across locations. Success rates also varied by orbital destination, even when accounting for differences in payload mass. Overall, mission reliability has shown a steady improvement since 2013.
- One hot encoding was applied to categorical variables in preparation for the machine learning model.

Notebook GitHub URL: [IBM-Data-Science-Capstone/Data Visualization.ipynb at main · elgels/IBM-Data-Science-Capstone]

EDA using Python's SQLite3

- According to Python documentation, SQLite3 is "a C library that provides a lightweight disk-based database that doesn't require a separate server process and allows accessing the database using a nonstandard variant of the SQL query language."*

- Python's SQLite3 module allows users to:
  - Connect to a database
  - Create a table
  - Execute SQL commands
  - Run SQL queries

- We used the Python's SQLite3 module in a Jupyter Notebook to to find payload information such as the total payload carried by boosters and the names of the boosters which have carried the maximum payload mass.

* sqlite3 — DB-API 2.0 interface for SQLite databases — Python 3.14.3 documentation

Notebook GitHub URL: IBM-Data-Science-Capstone/EDA using SQL.ipynb at main · elgels/IBM-Data-Science-Capstone

Building Interactive Maps with Folium

- According to Folium documentation: “Folium allows users to create interactive maps and geospatial visualizations that can be shared as standalone HTML files or integrated into web applications…Folium is a Python map plotting library that allow users to create interactive maps with features such as zoom-in, zoom-out, and the ability to easily find any coordinates on the map.”*

- We created site maps using circles, color-labeled markers and icons to identify the launch sites with the highest success rates.

- We analyzed launch site geographical locations and their proximities to railways, highways, coastline and cities.

* Folium Documentation

Notebook GitHub URL: IBM-Data-Science-Capstone/Launch Sites Locations Analysis with Folium.ipynb at main · elgels/IBM-Data-Science-Capstone

Building a Dashboard with Plotly Dash

- According to the Plotly Dash documentation: “Plotly Dash is an open-source Python framework for building analytical web applications and interactive data dashboards using pure Python.”*

- We developed an interactive dashboard featuring a drop-down menu and a payload range slider to explore launch performance. The dashboard included a pie chart that displayed the share of successful launches across all sites, with the option to select a specific site for more detailed insights.

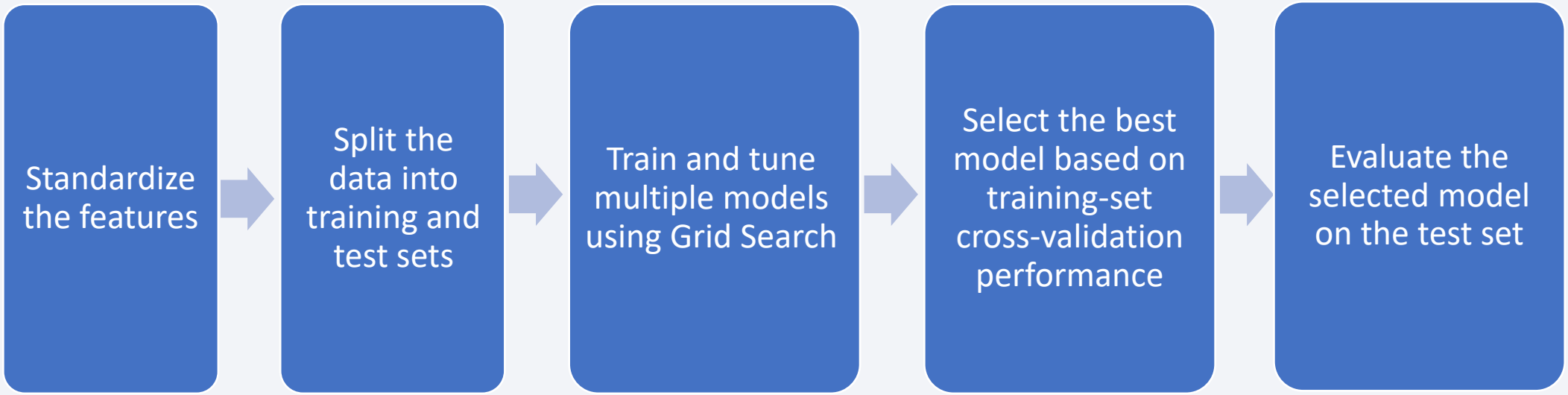
- It also incorporated a scatter plot showing the relationship between payload mass and launch outcome for all sites. The payload slider allowed users to filter the data by mass range, and booster versions were represented with distinct colors to highlight performance differences.

* Dash Documentation & User Guide | Plotly

Notebook GitHub URL: IBM-Data-Science-Capstone/Build a Dashboard Application with Plotly Dash.py at main · elgels/IBM-Data-Science-Capstone

Predictive Analysis (Classification)

- We created the variable “Class” (1 if launch was successful, 0 if launch failed) as the dependent variable. The variables Date, Outcome, Booster Version, Longitude and Latitude were excluded from the model.
- We applied hot encoding to categorical variables.
- Machine learning pipeline steps:



Notebook GitHub URL: [IBM-Data-Science-Capstone/Machine Learning Prediction.ipynb at main · elgels/IBM-Data-Science-Capstone]

Section 2

Insights drawn from EDA

Falcon 9 Launches and Landing Outcomes

- 61% of launches originated in Cape Canaveral.

CCAFS SLC (Cape Canaveral)	KSC LC 39A (Kennedy Space Center)	VAFB SLC 4E (Vandenberg)
55	22	13

- 70% of rockets launched in low Earth orbit.

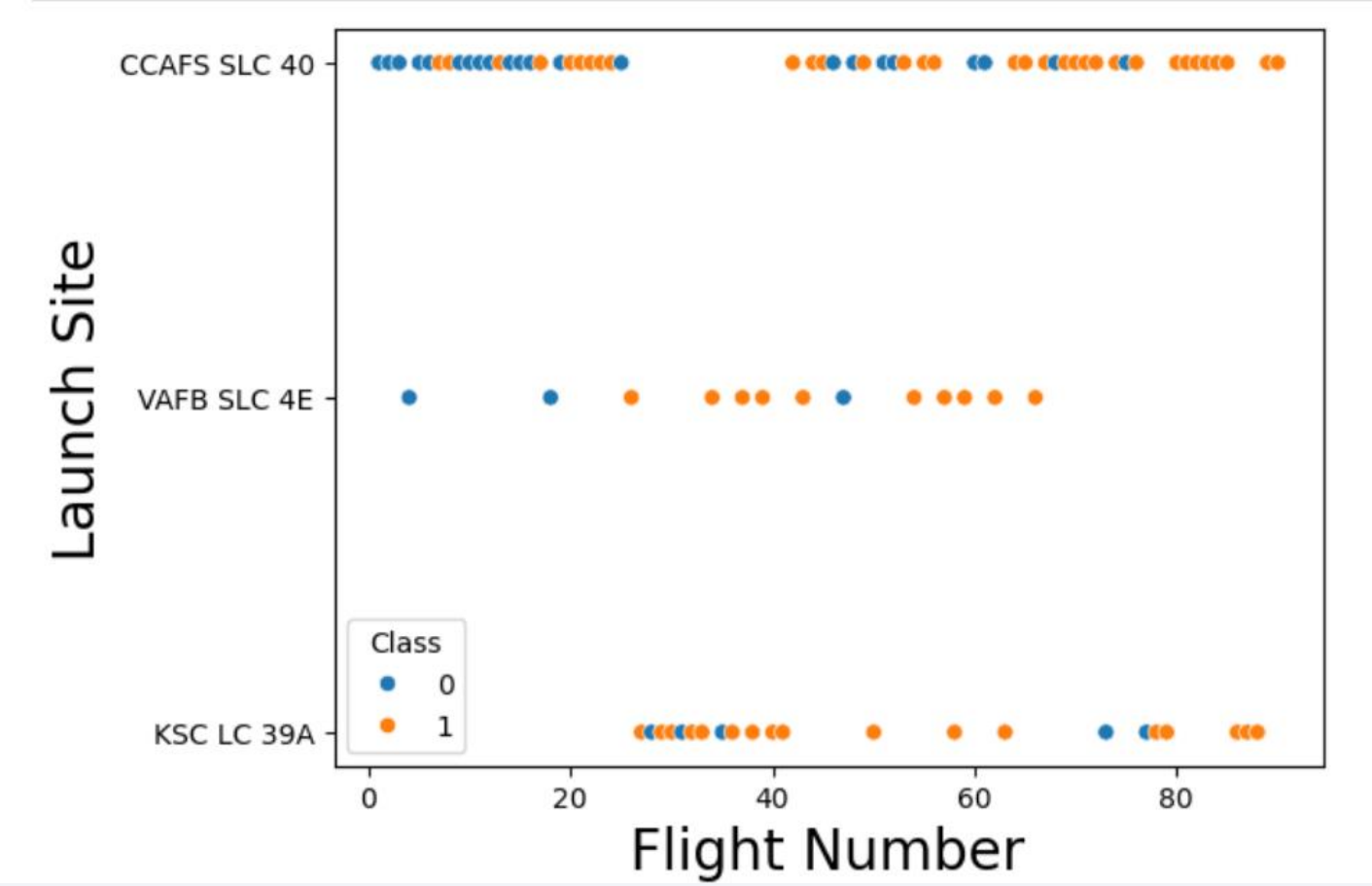
GTO	ISS	VLEO	PO	LEO	Other
27	21	14	9	7	12

- 67% of launches landed successfully in a drone ship or a ground pad.

True ASDS (Successfully landed on a drone ship)	None (Failure to Land)	True RTLS (Successfully landed to a ground pad)	False ASDS (Unsuccessfully landed on a drone ship)	True Ocean (Successfully landed in the ocean)	Other
41	19	14	6	5	5

- New variable “Class” used column Outcome as source to determine success (1) or failure (0).

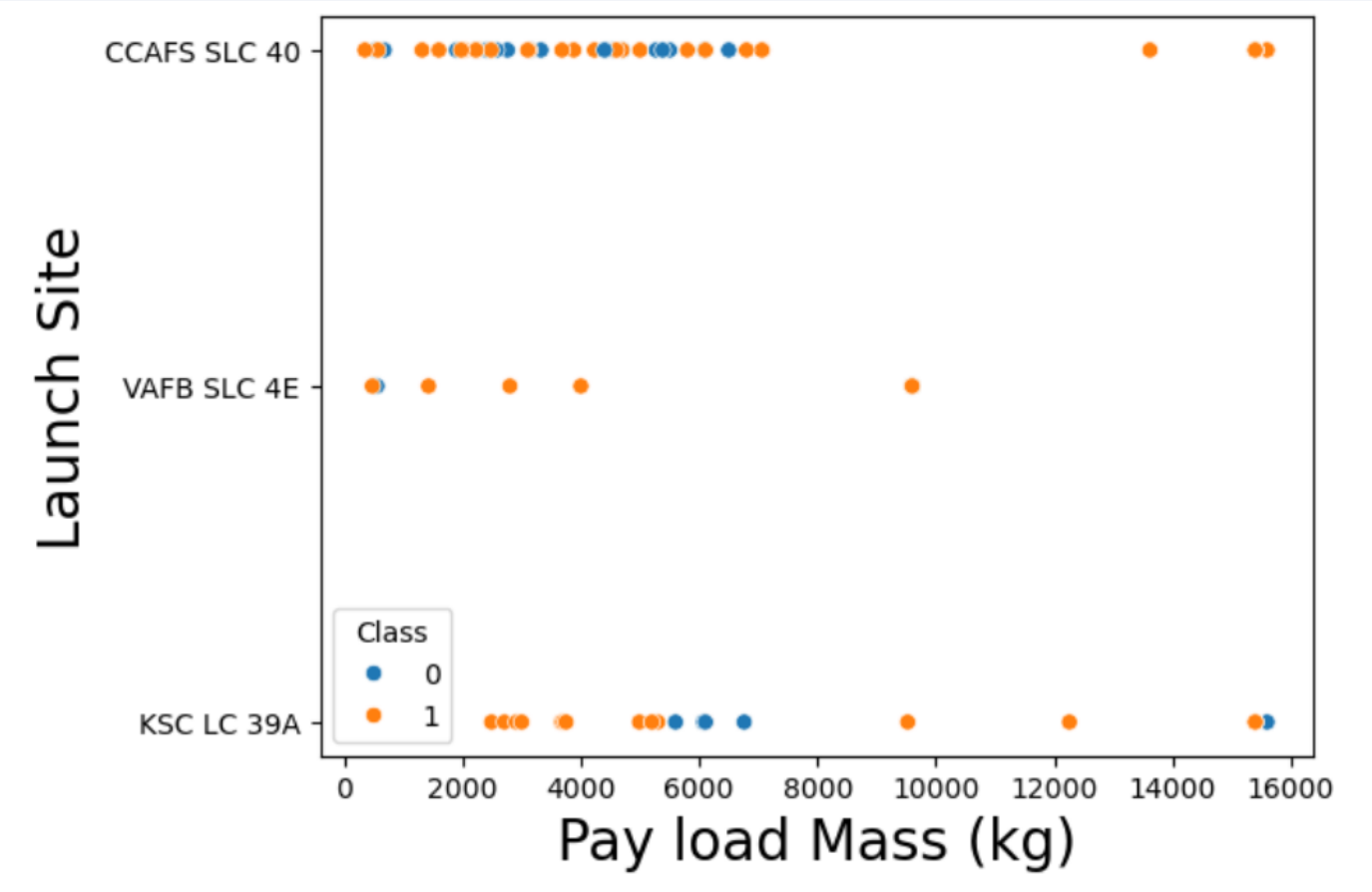
Number of Flights and Launch Site



- Higher number of flights seem to correspond with more successful launches at all sites.
- The launch site VAFB SLC (Vandenberg) did not have more flights beyond the 70th flight mark.

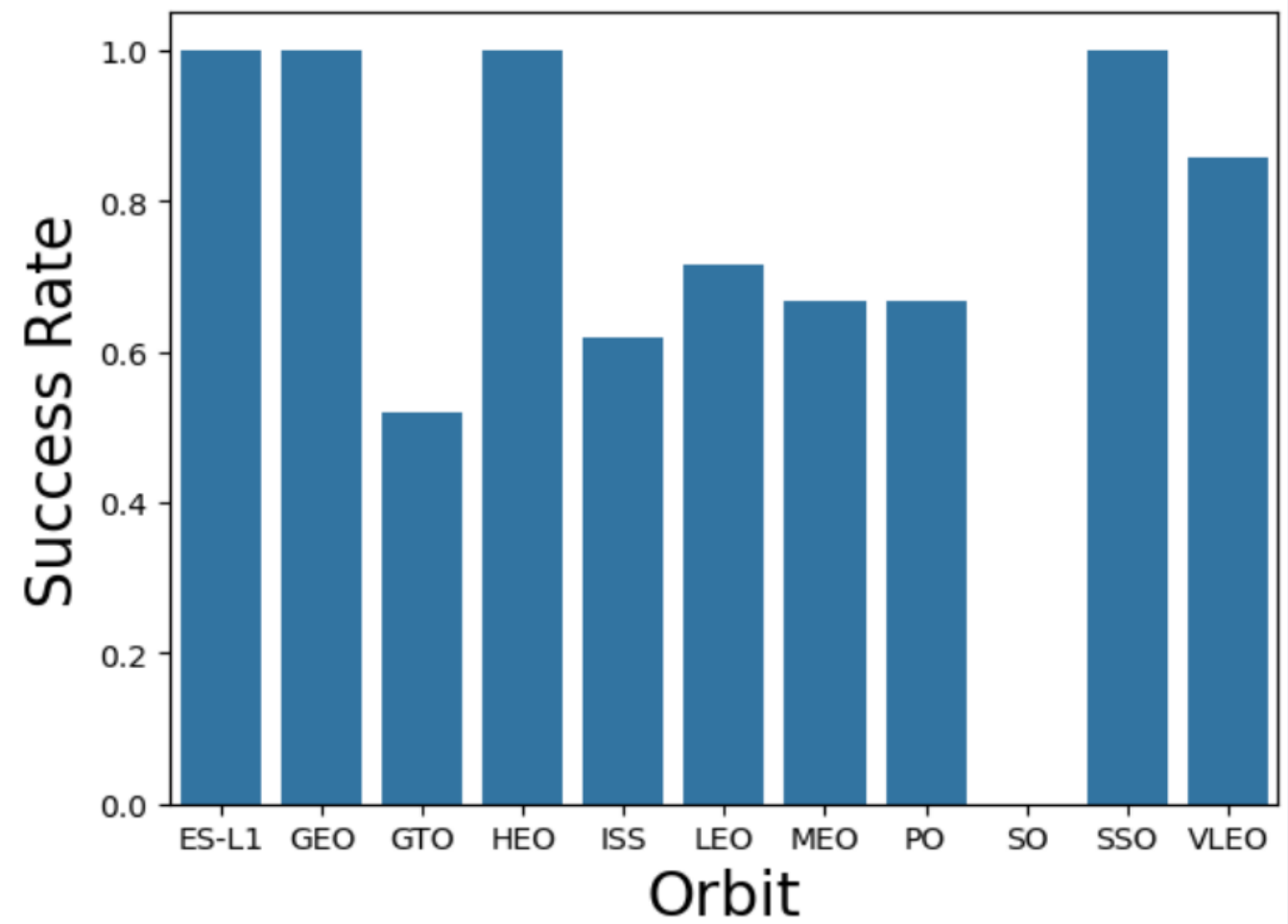
Payload (kg) and Launch Site

- At the CCAFS SLC 40 launch site (Canaveral Center), most rockets carried up to 8,000 kg; but with mixed success rates.
- The VAFB-SLC (Vandenberg) launch site had no rockets launched for heavy payload mass (greater than 10,000 kg).



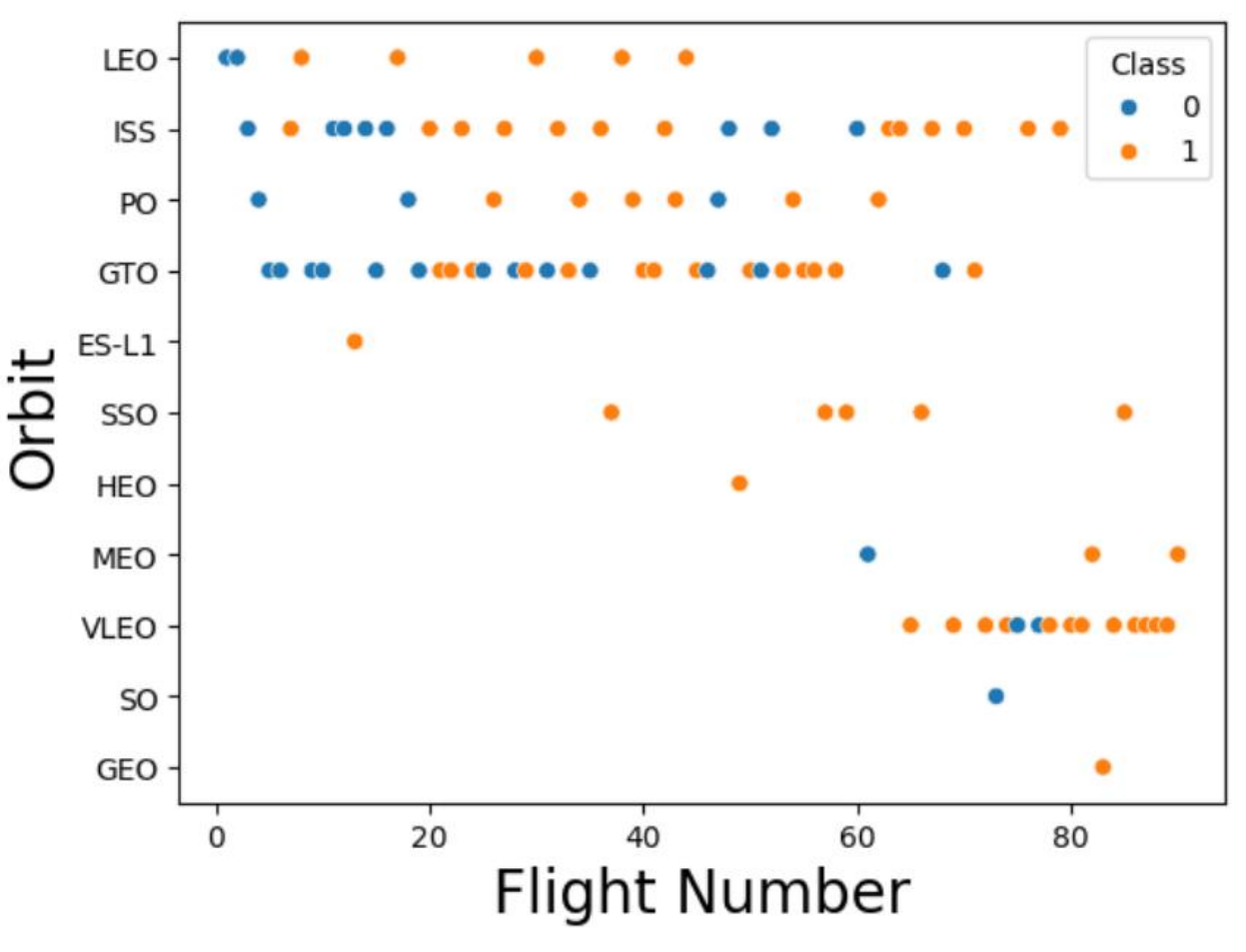
Success Rate and Orbit Type

- The Falcon 9 had high success rates in low Earth orbits.
- ES-L1, HEO, SSO and GEO had one to five flights only and they were all successful.
- VLEO: Very low Earth orbit had the highest success rate for orbits with 10 launches or more.



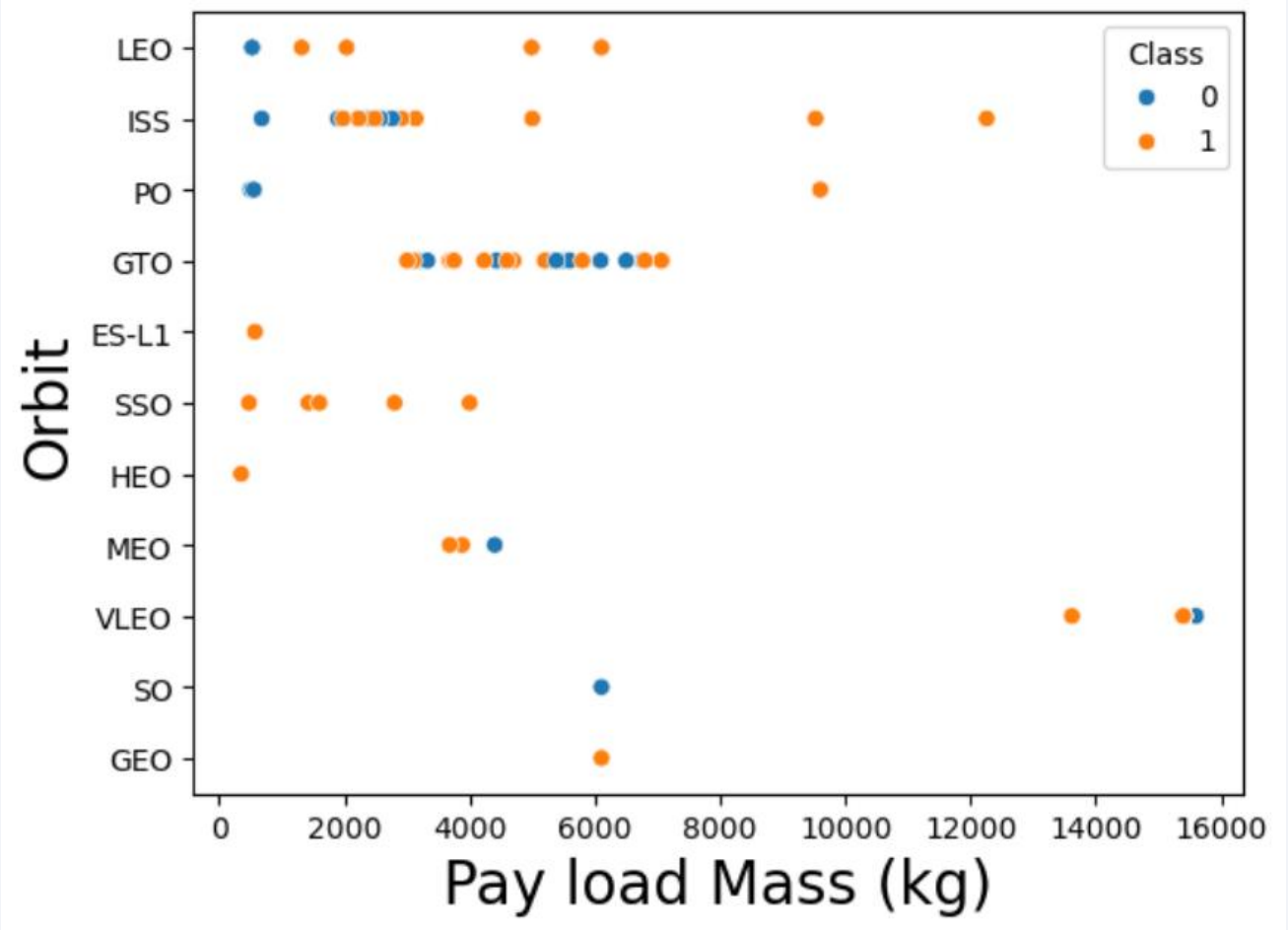
Number of Flights and Orbit Type

- In the LEO orbit, the success rate appears to increase with the number of flights, but in GTO there seems to be no relationship between them.
- The VLEO orbit had mostly successes for any number of flights.



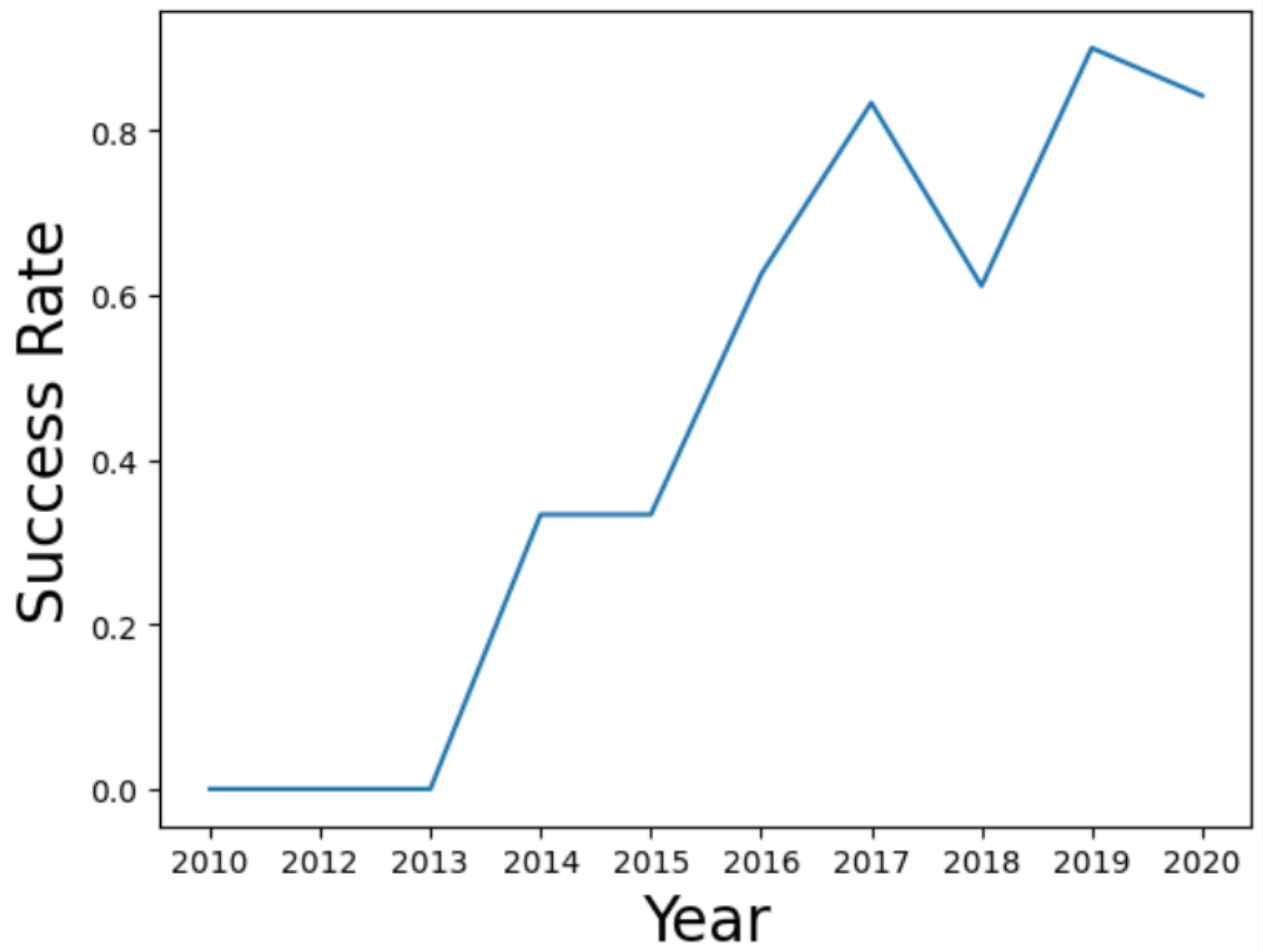
Payload (kg) and Orbit Type

- It does not seem to be a clear correspondence between payload and orbit type.
- For GTO specially, payload and success rates are not related.
- SSO had all successful launches (five) under 4,000 kg of payload mass.



Launch Success Yearly Trend

- The success rate has been increasing since 2013.
- There was a sudden dip in success rates in 2018. That year there were two failures to land: One in February (drone ship) and one in December (ground pad).



SQLite3: Launch Sites

SQL Query:	%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
Result:	Launch Sites: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

- Cape Canaveral Space Launch Complex (CCAFS SLC-40); and previously Launch Complex 40 (CCAFS LC-40) is located at the north end of Cape Canaveral Space Force Station, Florida.
- Kennedy Space Center (KSC LC 39A) is located in Merritt Island, Florida.
- Vandenberg Space Force Base (VAFB SLC-4E) is located in Santa Barbara County, California.

SQLite3: Payload Mass (kg)

SQL Query:	%sql SELECT customer, SUM(PAYLOAD_MASS__KG_) AS total_payload FROM SPACEXTABLE WHERE customer="NASA (CRS)" GROUP BY customer;
Result:	Customer: NASA (CRS) Sum of Payload: 45,596

SQL Query:	%sql SELECT "Booster_Version", AVG(PAYLOAD_MASS__KG_) AS avg_payload FROM SPACEXTABLE WHERE "Booster_Version"="F9 v1.1" GROUP BY "Booster_Version"
Result:	Booster Version: F9 v1.1 Average Payload: 2,928.4 kg

- The total payload carried by boosters belonging to customer NASA (CRS) was 45,596 kg.
- The average payload mass carried by booster version F9 v1.1 was 2,928.4 kg.

SQLite3: Payload Mass (kg)

SQL Query:	%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing Outcome"="Success (drone ship)" AND (PAYLOAD_MASS__KG_>4000 AND PAYLOAD_MASS__KG_<6000) ;
Result:	Booster Version: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2

SQL Query:	%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) AS max_payload FROM SPACEXTABLE);
Result:	F9 B5 B1048.4, F9 B5 B1049.4, F9 B5 B1051.3, F9 B5 B1056.4, F9 B5 B1048.5, F9 B5 B1051.4, F9 B5 B1049.5, F9 B5 B1060.2, F9 B5 B1058.3, F9 B5 B1051.6, F9 B5 B1060.3, F9 B5 B1049.7

- The names of the booster versions that had success in drone ship landing and a payload mass greater than 4,000 kg but less than 6,000 kg, were: F9 FT B1022, F9 FT B1026, F9 FT B1021.2 and F9 FT B1031.2.
- The booster versions which have carried maximum payload mass were: F9 B5 B1048.4, F9 B5 B1049.4, F9 B5 B1051.3, …and F9 B5 B1049.7.

SQLite3: Successful Outcomes

SQL Query:	%sql SELECT "Mission_Outcome", COUNT(*) AS count_misssion_outcome FROM SPACEXTABLE GROUP BY TRIM("Mission_Outcome");
Result:	Success: 99 Failure (in flight): 1 Success (payload status unclear): 1

SQL Query:	%sql SELECT "Landing Outcome", MIN("Date") AS min_date FROM SPACEXTABLE WHERE "Landing Outcome"="Success (ground pad)" GROUP BY "Landing Outcome";
Result:	Landing Outcome: Success (ground pad) Minimum Date: 2015-12-22

- The total number of successful and failure mission outcomes were:
  - Success: 99
  - Failure (in flight): 1
  - Success (payload status unclear): 1

- The first successful landing outcome on ground pad was on December 22nd, 2015.

SQLite3: Landing Outcomes by Date

SQL Query:	%sql SELECT "Landing_Outcome", Count("Landing_Outcome") AS count_landing_outcomes FROM SPACEXTABLE WHERE ("Landing_Outcome" = "Failure (drone ship)" OR "Landing_Outcome" = "Success (ground pad)") AND ("Date" BETWEEN "2010-06-04" AND "2017-03-20") GROUP BY "Landing_Outcome" ORDER BY "count_landing_outcomes" DESC;
Result:	Count of Landing Outcomes: 5, 3

- The number of landing outcomes with Failure (drone ship) or Success (ground pad); between the date 2010-06-04 and 2017-03-20 was:

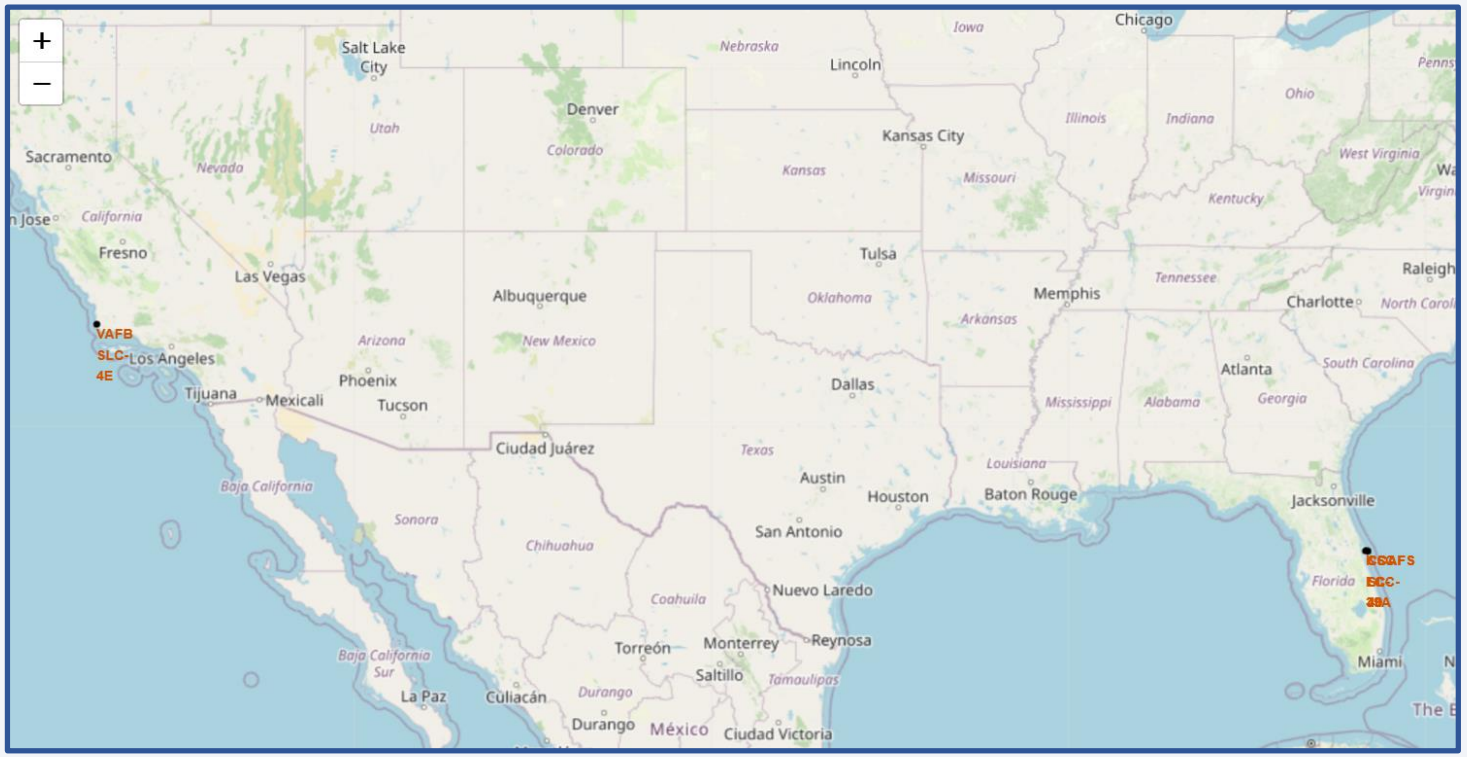
Landing Outcome	Count
Failure (drone ship)	5
Success (ground pad)	3

Section 3

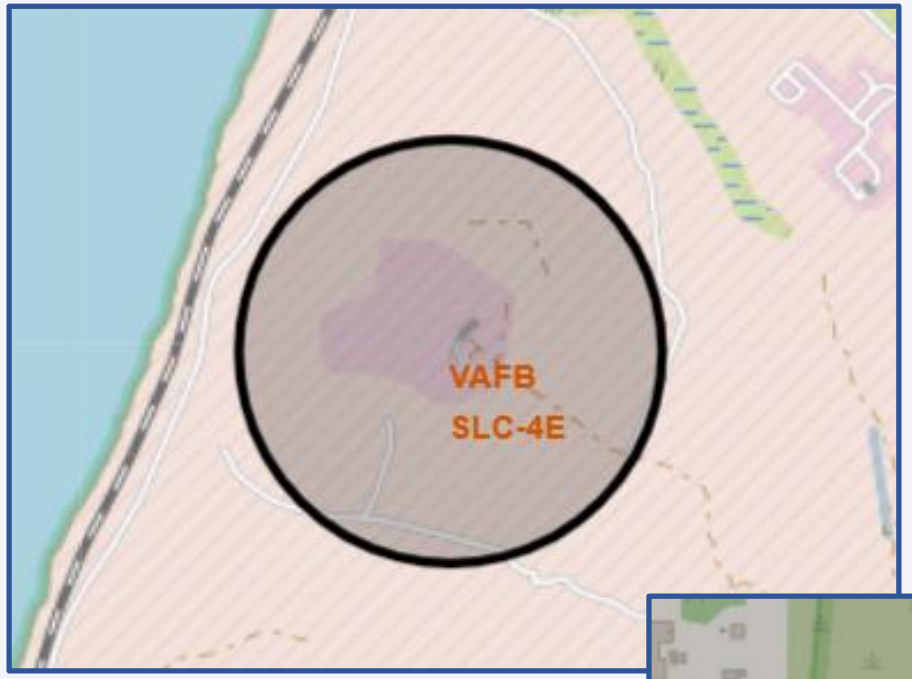
Launch Sites Proximities Analysis

Launch Sites Map

- The launch sites in Florida (CCAFS SLC-40, KSC LC 39A) are closer to the Equator line that the launch site in California (VAFB SLC-4E).
- All launch sites are close to the Ocean. In Florida, the launch sites are located by the Atlantic Ocean and in Los Angeles, the site is close to the Pacific Ocean.



Launch Sites Map - Close-Up



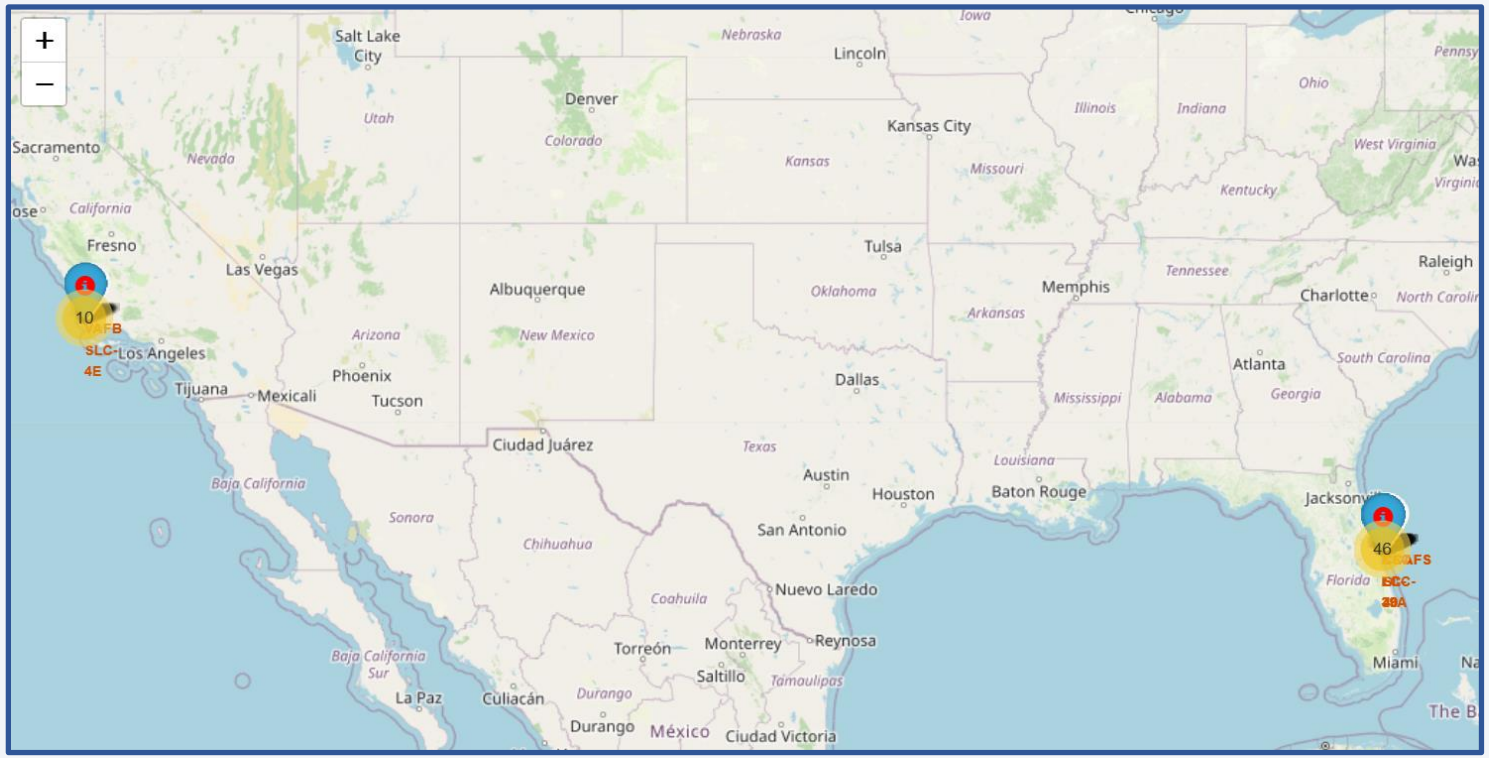
Vandenberg Space Force Base, California.

Cape Canaveral Space Force Station, Florida.

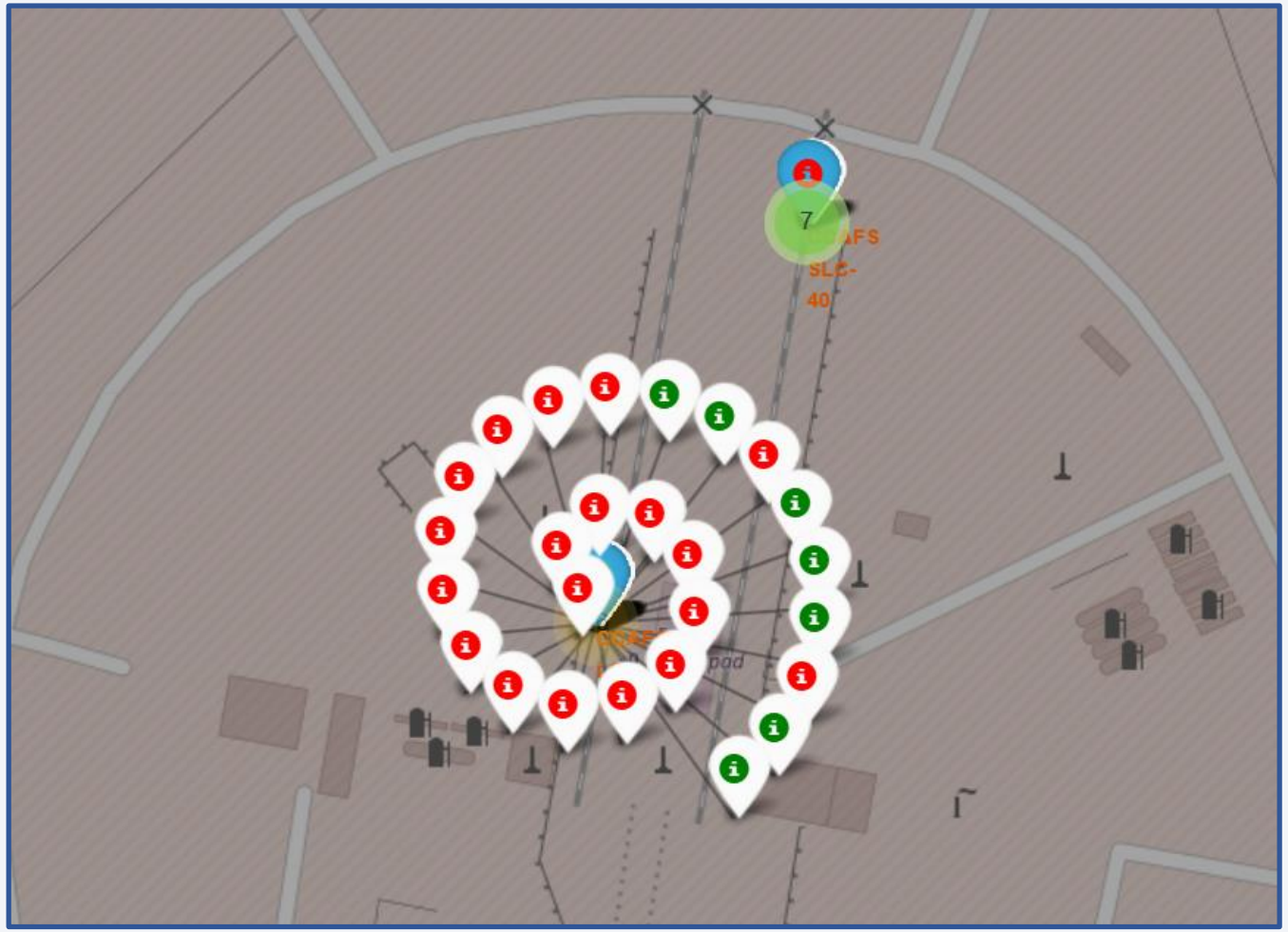
Kennedy Space Center, Florida.

Launch Sites Map and Outcomes

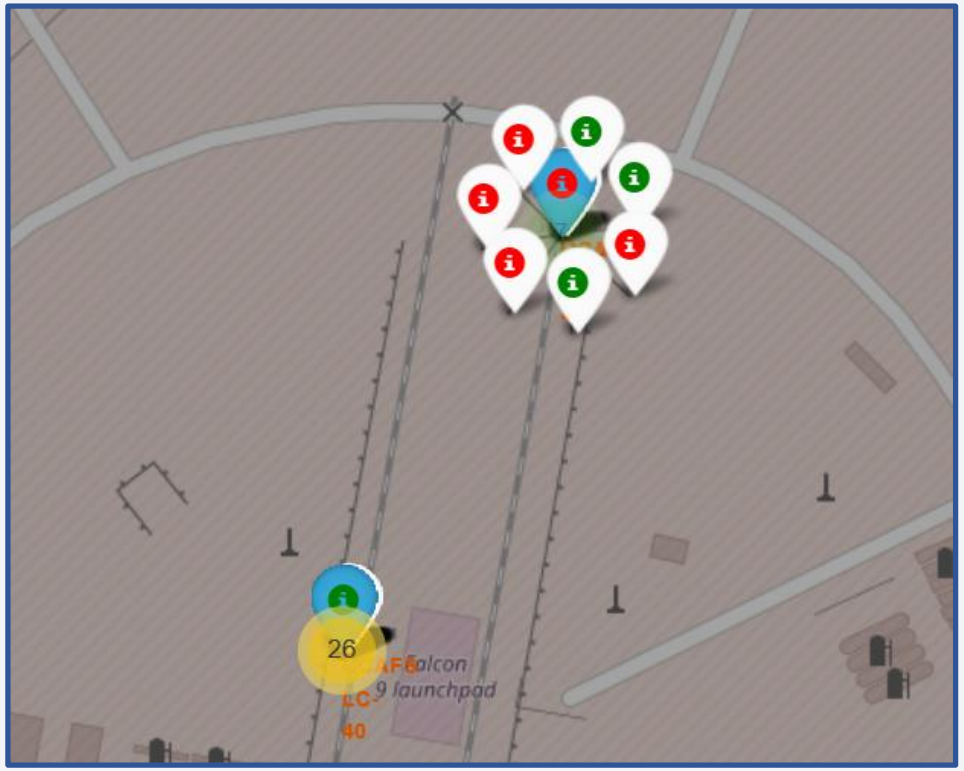
- This map shows the launch outcomes for each site. A green marker means success and a red marker means failure.
- The Kennedy Space Center (KSC LC-39A) in Florida had the highest success rate.



Launch Sites Map and Outcomes - Close-Up 1



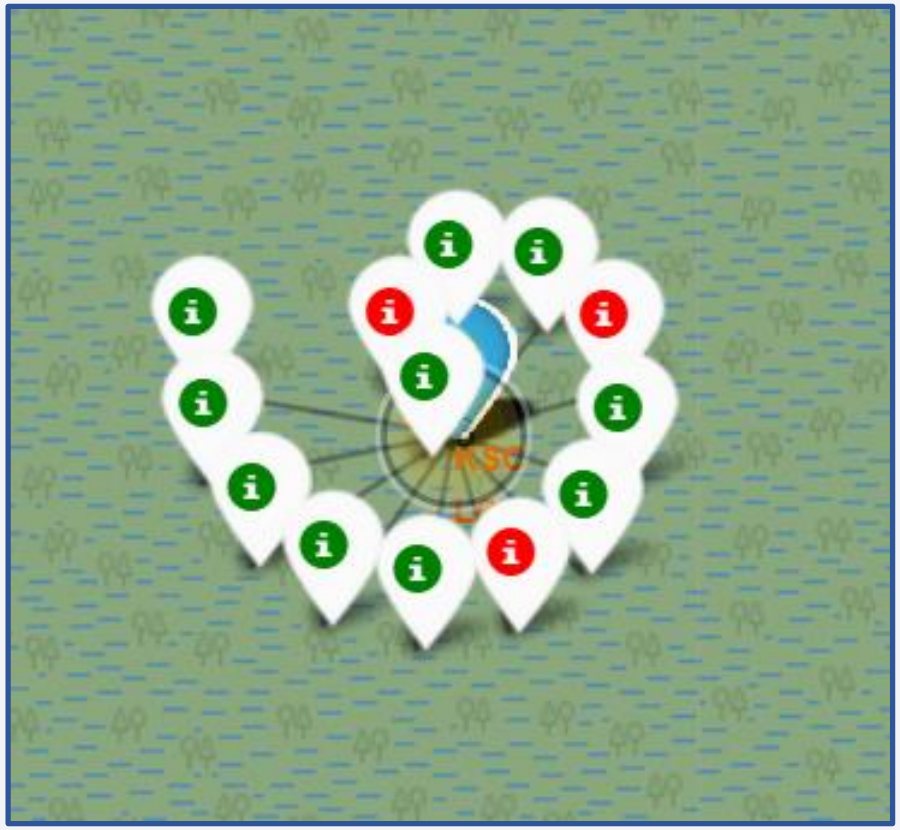
CCAFS LC -40



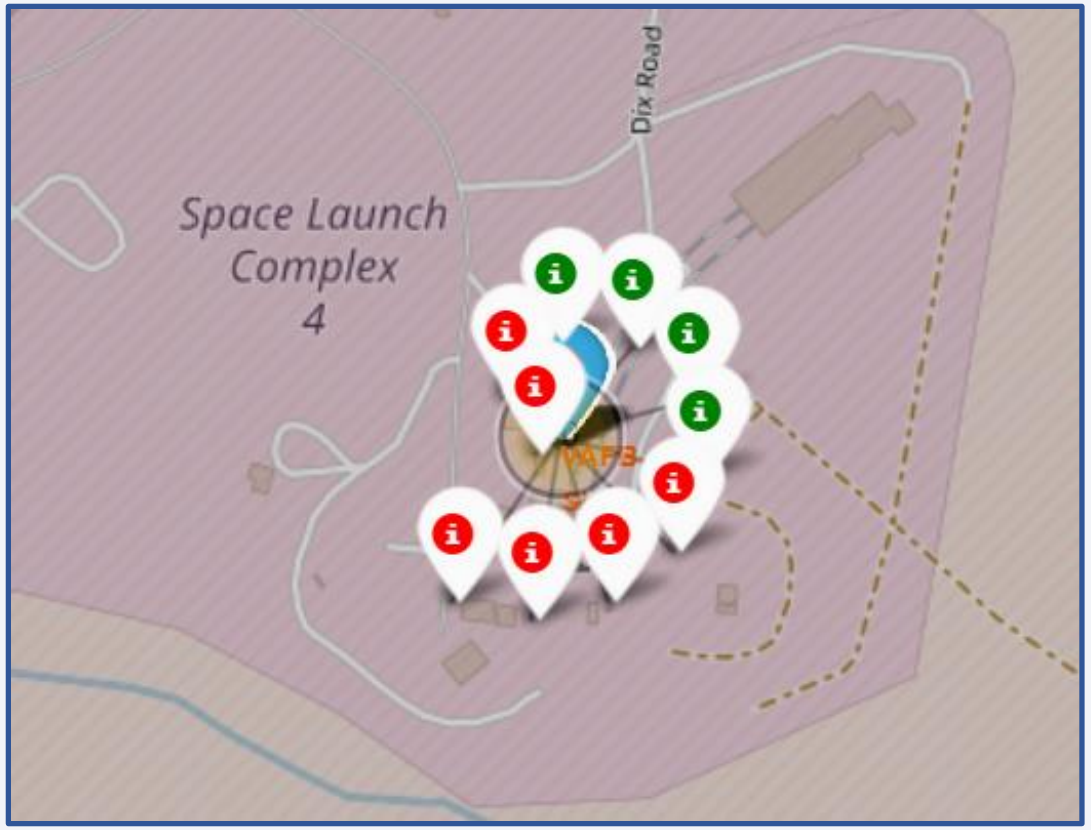
CCAFS SLC -40

Cape Canaveral Space Force Station, Florida.

Launch Sites Map and Outcomes - Close-Up 2

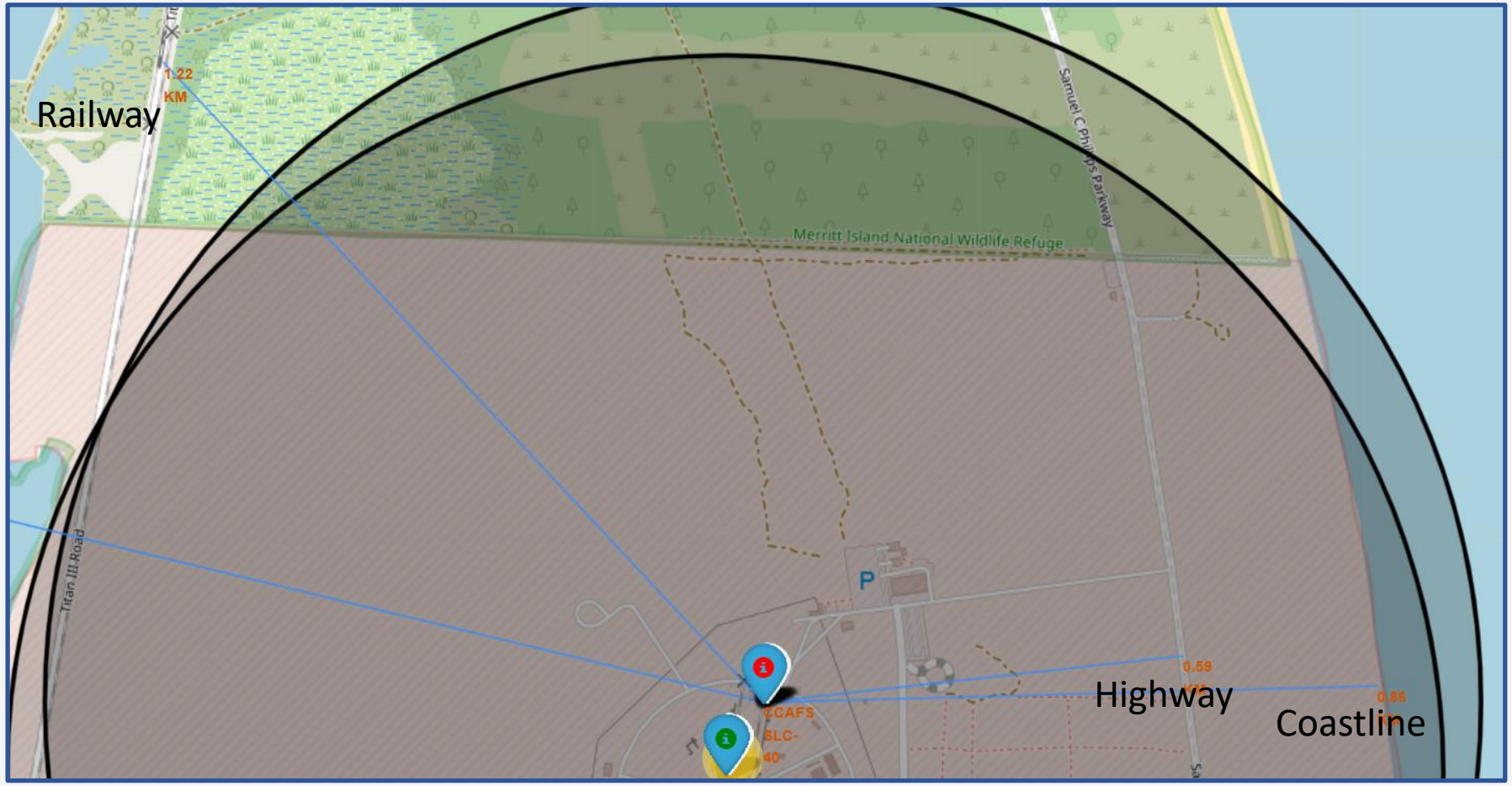


Kennedy Space Center
(KSC LC-39A), Florida.



Vandenberg Space Force Base
VAFB SLC-4E , California.

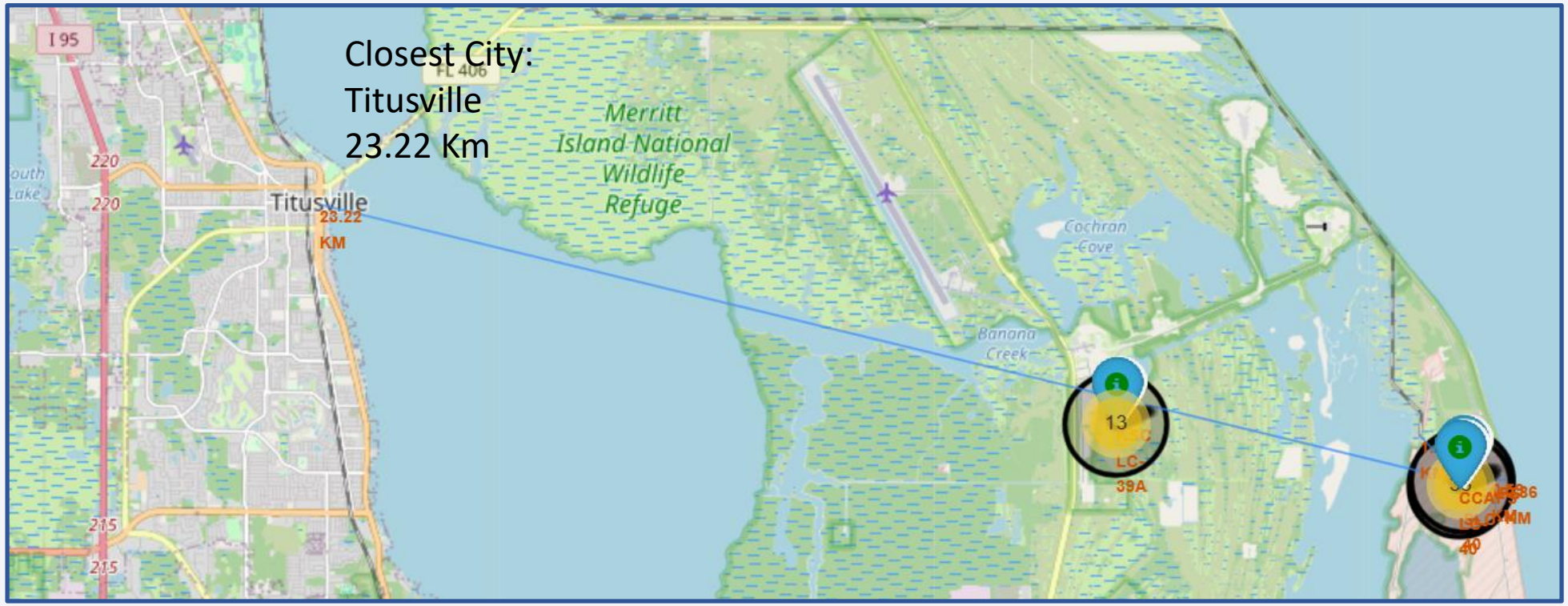
Canaveral Site and its Closest Railway, Highway and Coastline



This map shows the Cape Canaveral Space Force Station (CCAFS SLC -40) and its proximities to:

- Railway: 1.22 Km
- Highway: 0.59 Km
- Coastline: 0.86 Km

Canaveral Site and its Closest City



The Cape Canaveral Space Force Station (CCAFS SLC -40) is 23.22 km away from its closest city (Titusville).

Proximity to Railways, Highways, Coastline and Cities

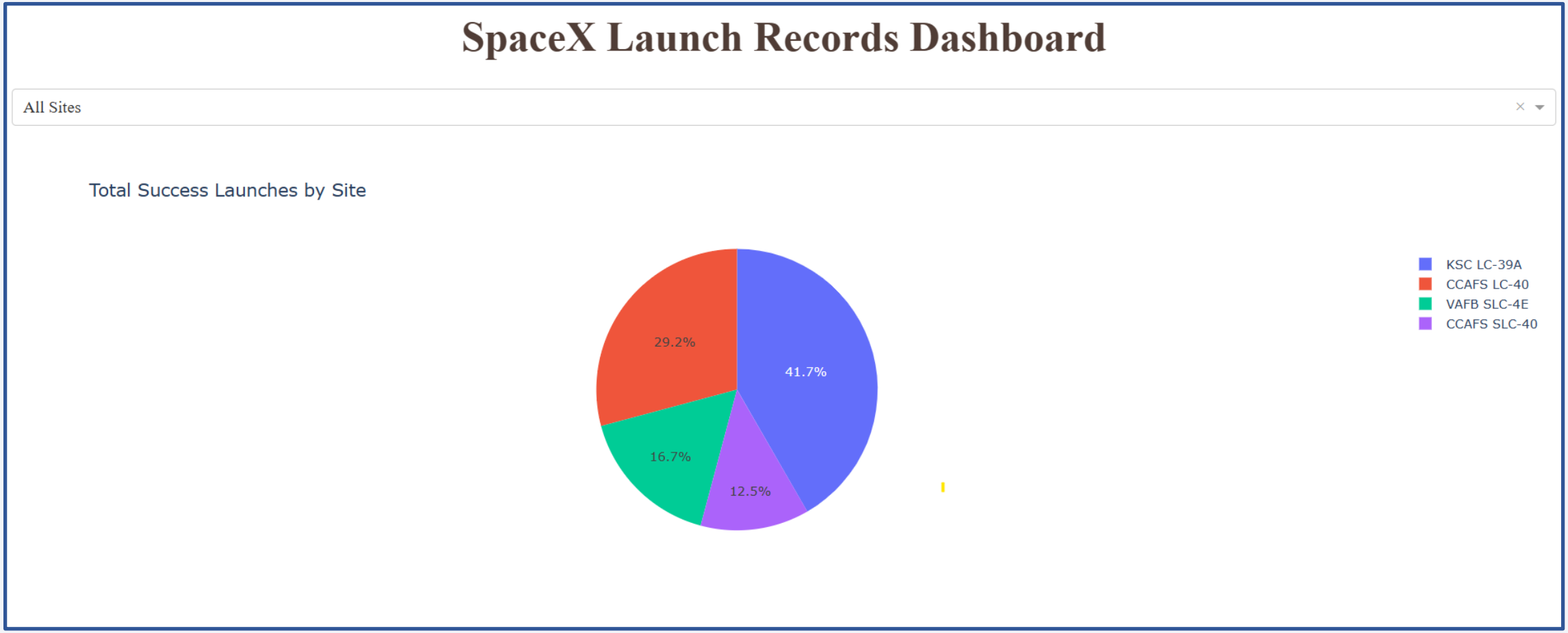
- Launch sites are in close proximity to railways and highways because it makes transportation of heavy equipment and materials more manageable. They also help maintain fuel efficiency.
- All launch sites are in close proximity to coastlines beacuse in case of failure to land, the rocket does not fall inland.
- All launch sites keep a good distance from cities because if something goes wrong during the ascent, the debris will not fall in a densely populated area.

Section 4

Build a Dashboard with Plotly Dash

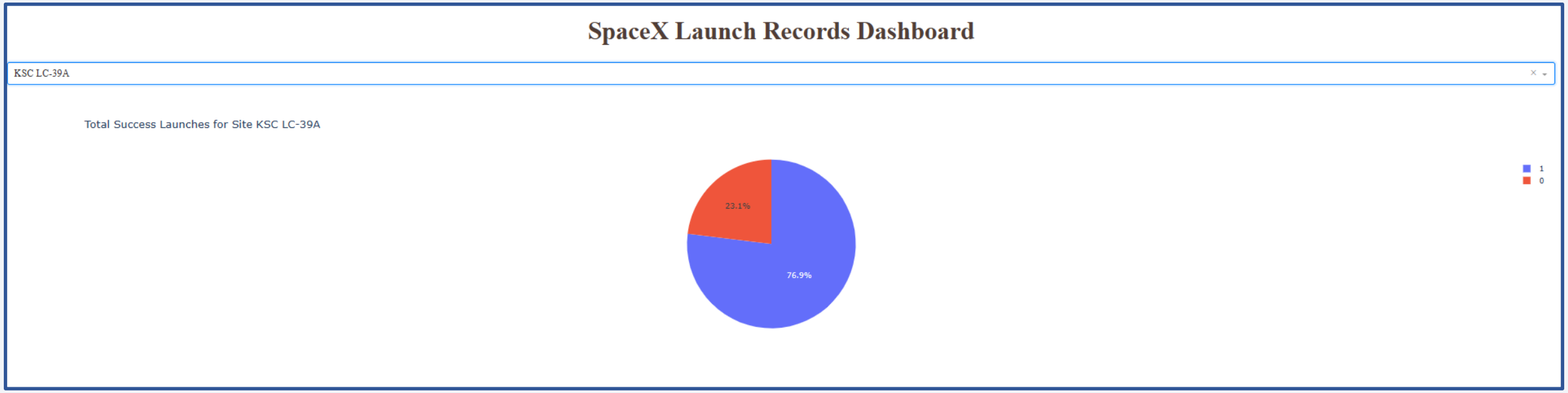
Success Rates Dashboard

- The Kennedy Space Center had the highest success rate (41.7%). The Canaveral Launch Site followed with 29.2%.



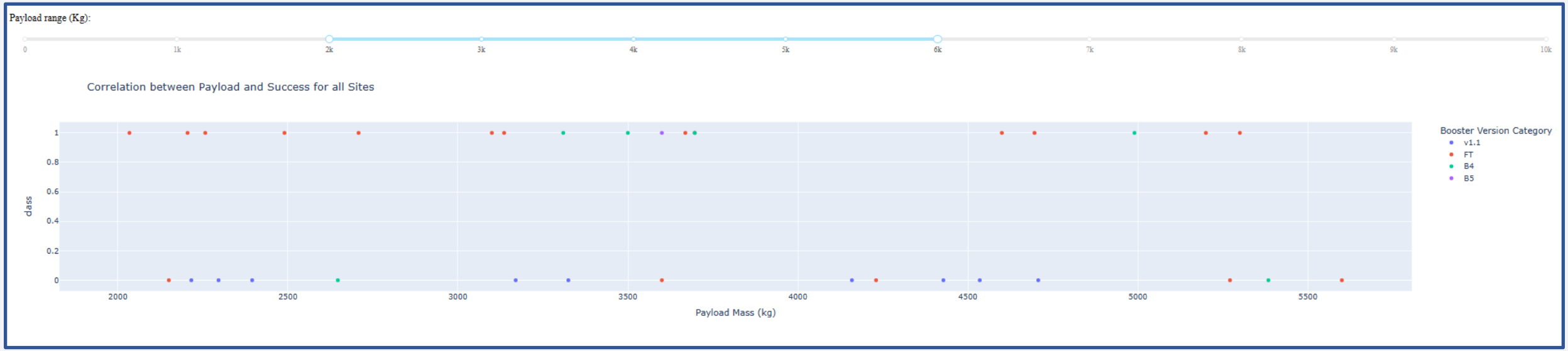
Success Rates Dashboard - Kennedy Space Center

- For the Kennedy Space Center (the launch site with the highest success rate), 76.9% of their total launches were successful while 23.1% failed.



Payload vs. Launch Outcome Scatter Plot

- The scatter plot shows the successful launches (“Class” value of 1) and unsuccessful launches (“Class” value of 0) for all sites, by different payload ranges and booster version categories.
- The payload range can be selected in the range slider at the top of the graph; and the booster categories are represented in different colors for v1.1, FT, B4 and B5.

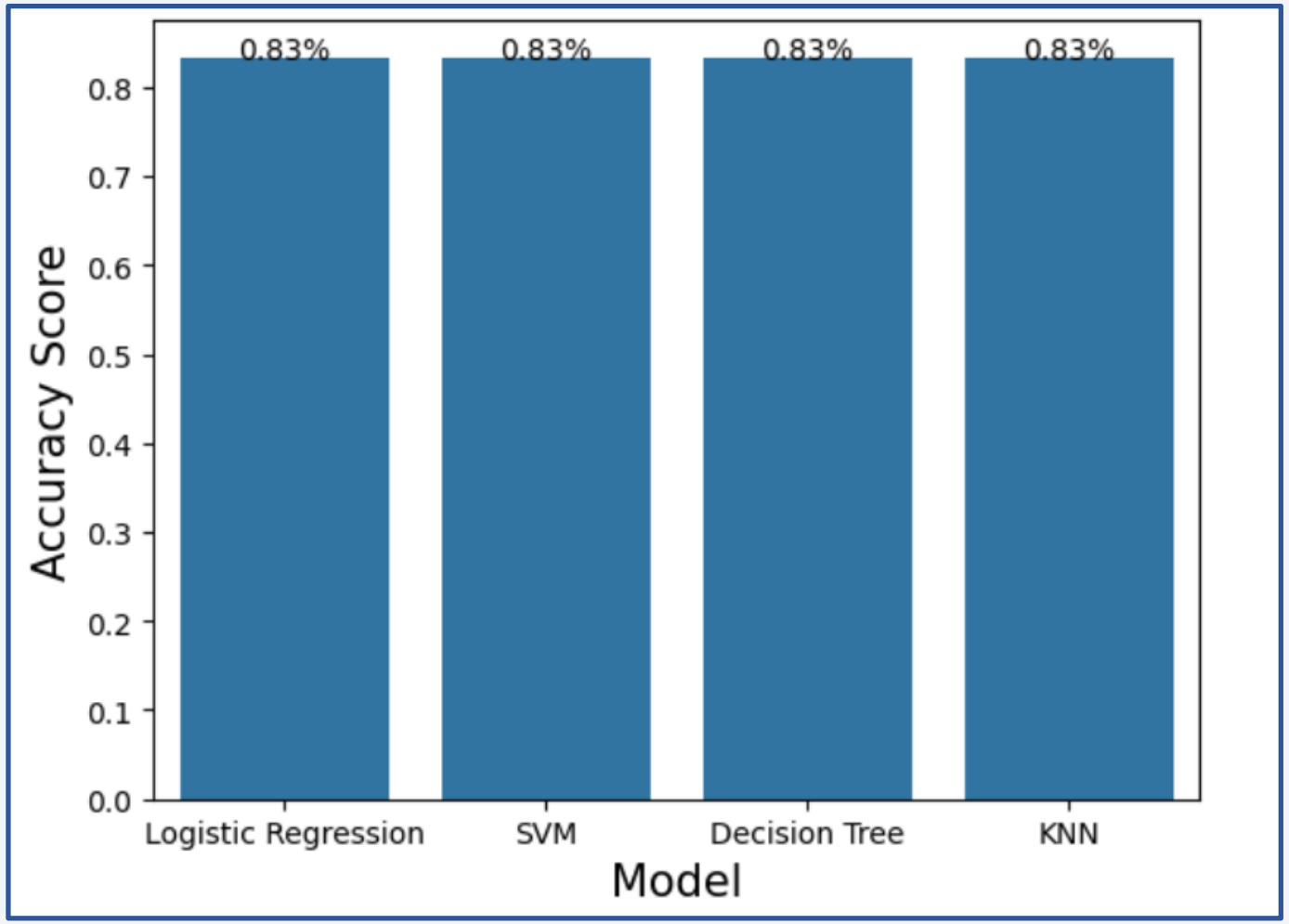


Section 5

Predictive Analysis (Classification)

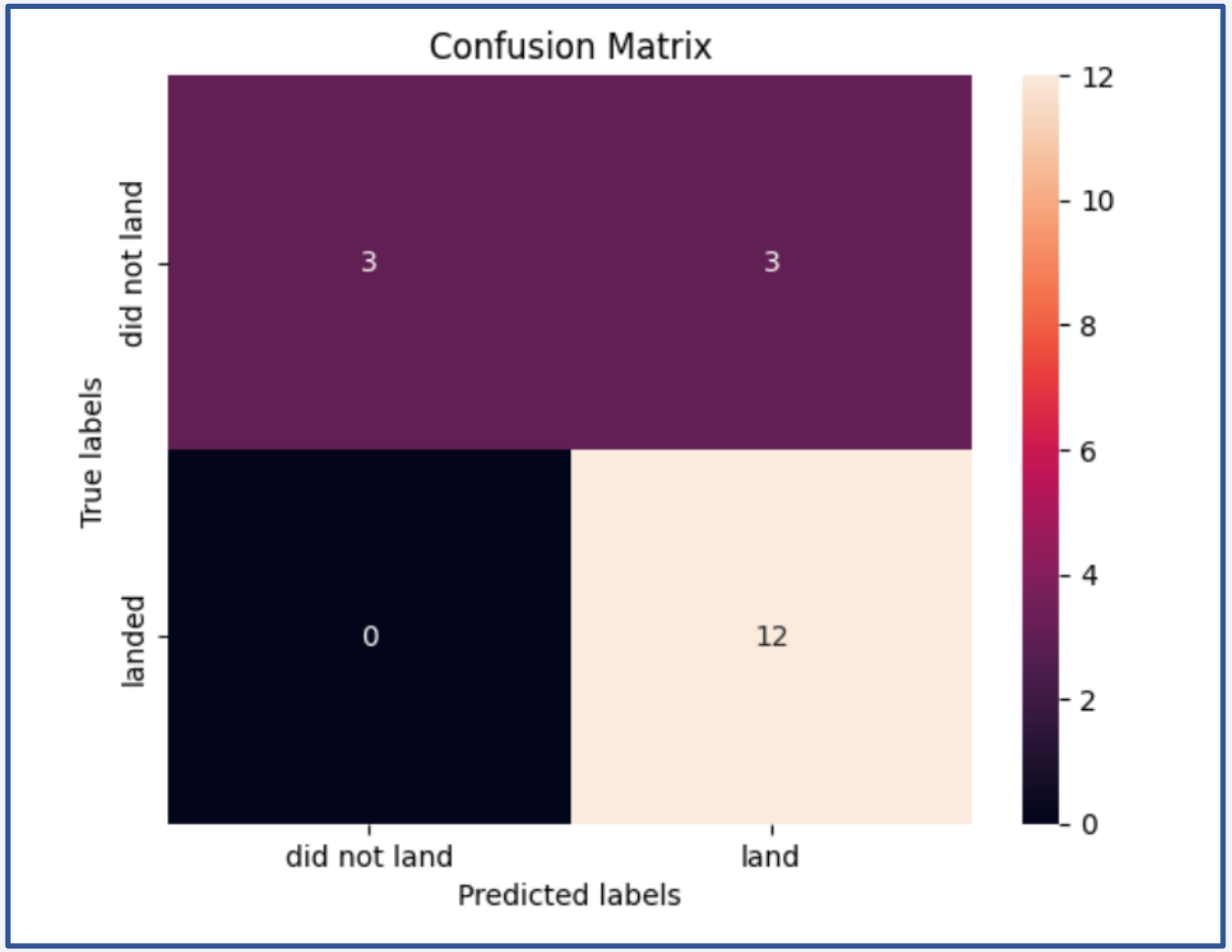
Classification Accuracy

- We developed and tested four predictive models:
  - Logistic Regression
  - Support Vector Machine
  - Decision Tree
  - K-Nearest Neighbors
- All four models had the same classification accuracy on the test data set: 0.83.



Confusion Matrix

- Since all four models had the same classification accuracy, they also had the same confusion matrix.
- We only had 18 observations for our dependent variable.
- True positives are high (12) meaning the model is good at predicting the launches that will be successful.
- False positives, meaning the model predicted a successful outcome (land) when in reality the launch failed (did not land), were 3.

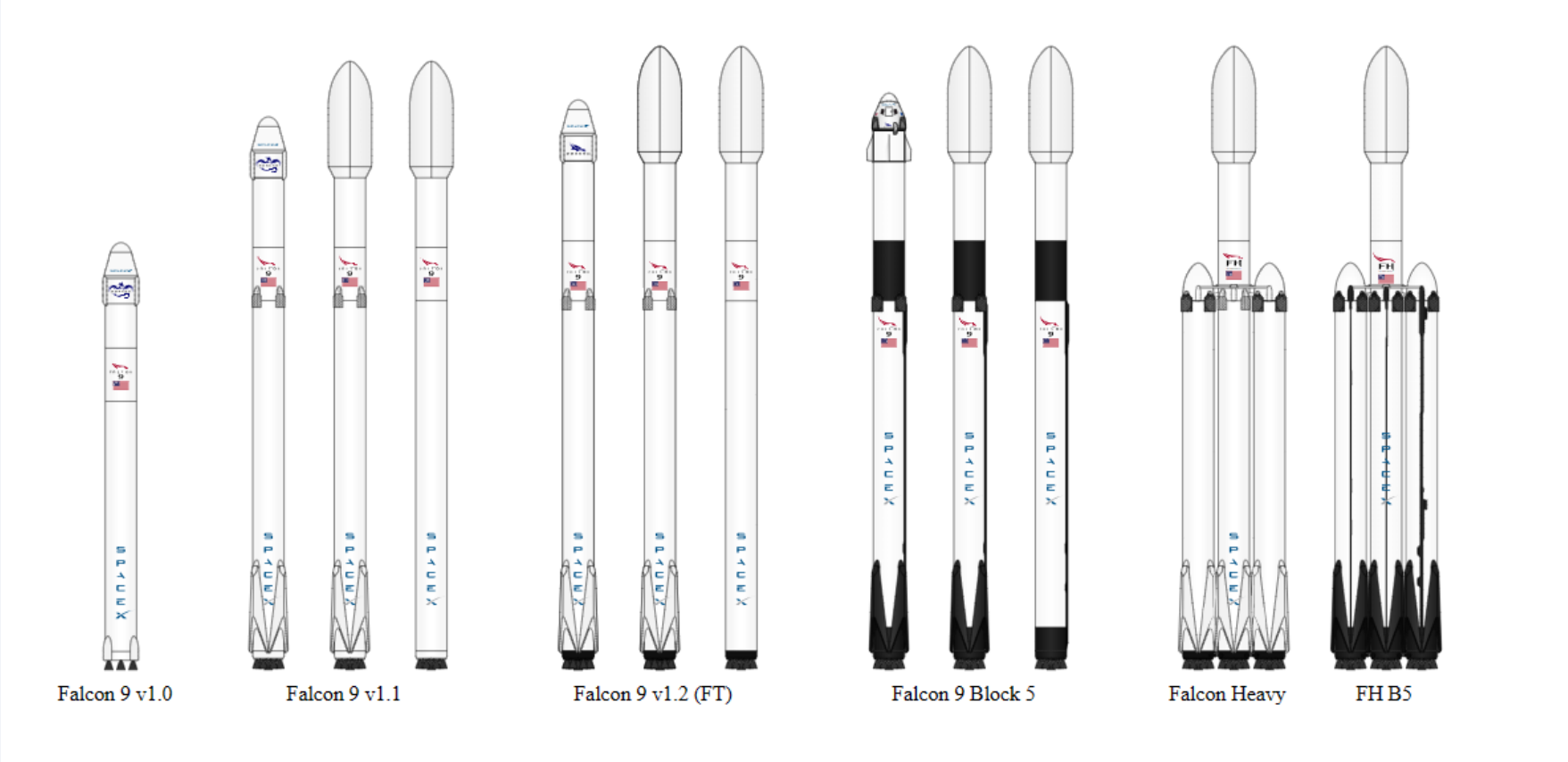


Conclusions

- The goal of this project was to develop a model capable of predicting whether the first stage of a Falcon 9 rocket would land successfully and be reused. Accurate predictions of landing outcomes can help estimate launch costs, enabling other companies to make more informed and competitive bids against SpaceX.
- Data for this analysis was collected through API requests and web scraping. The raw data was cleaned and wrangled into a structured format suitable for analysis. During exploratory data analysis, visualizations were used to examine relationships between variables and to identify relevant features for modeling.
- After preprocessing and standardizing the data, four classification models were developed and evaluated: Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN). All models achieved identical performance on the test set, with an accuracy of 0.833 and identical confusion matrices. This suggests that the dataset is relatively simple and that different algorithms are able to capture the underlying patterns equally well.
- Despite similar predictive performance, the models differ in interpretability and complexity. Logistic Regression offers a simpler and more interpretable approach, while Decision Trees and KNN may be more prone to overfitting. Therefore, Logistic Regression may be preferred as a baseline model due to its transparency and robustness.
- Future work could focus on incorporating additional variables that better capture operational, environmental, and technical factors influencing landing outcomes. Potential features include rocket trajectory parameters, weather conditions at launch and landing, records of technical anomalies, the vehicle's flight history, and overall space traffic conditions.

Appendix

Falcon 9 first-stage boosters

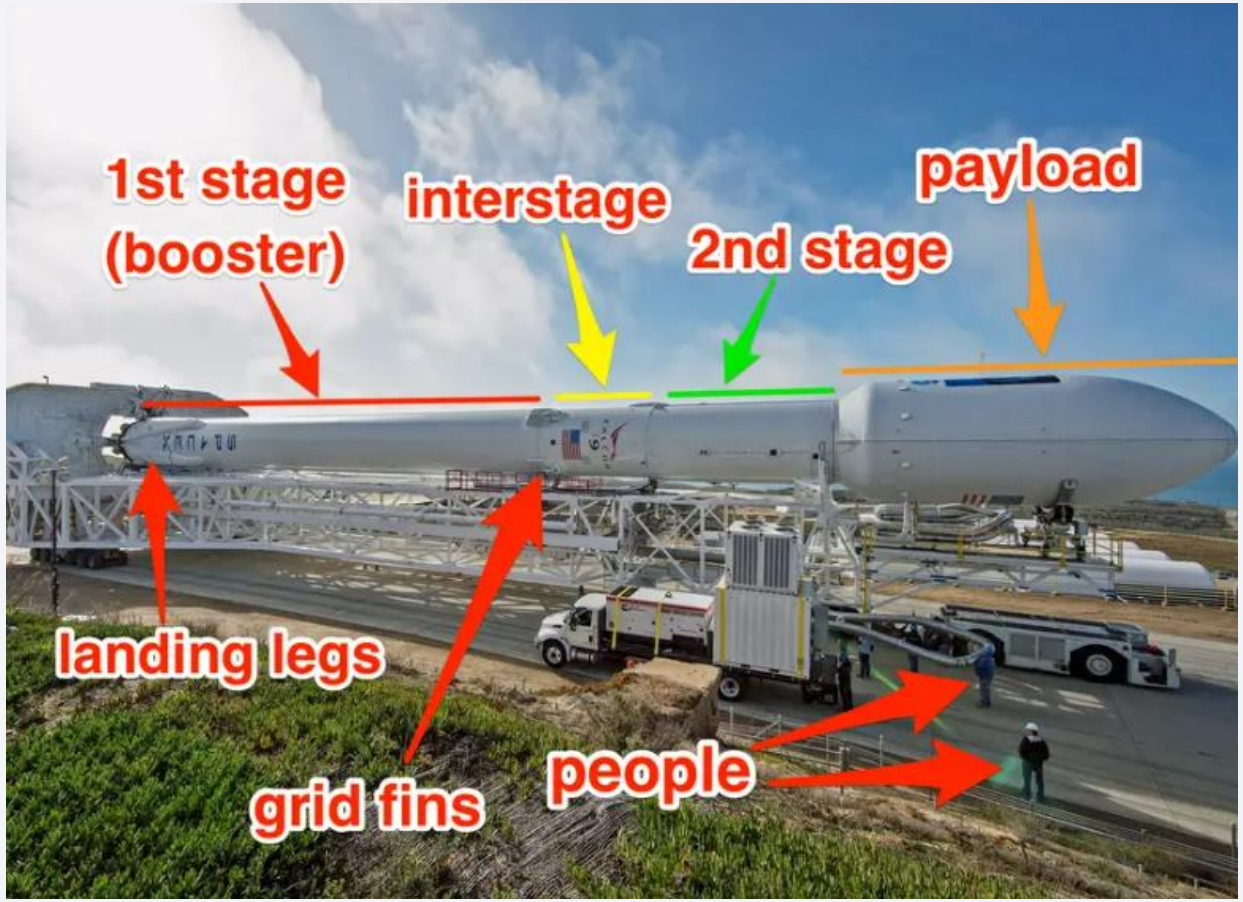


[Wikipedia Falcon 9]

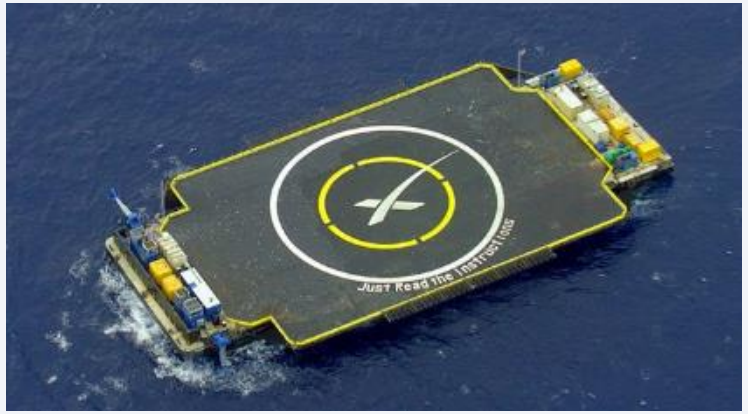
Appendix

- Falcon 9 is a rocket design that features reusable first-stage boosters. There are three booster types: Falcon 9 (F9), Falcon Heavy core (FH core), and Falcon Heavy side (FH side). Falcon 9 Boosters are numbered with a B followed by a four-digit number.
- The “FT” in Falcon 9 stands for Full Thrust, which refers to the upgraded version of the Falcon 9 rocket developed by SpaceX. The Full Thrust version, also known as Falcon 9 v1.2, introduced several improvements over its predecessors, including increased thrust, enhanced performance, and the capability for both stages of the rocket to be reusable.
- Our data ranged from June 2010 to November 2020. Now we know that in 2023 SpaceX was very successful: They performed 91 launches of Falcon 9 with only 4 utilizing new boosters and successfully recovered the booster on all flights.

Appendix



Note: The second stage, powered by a single Merlin Vacuum Engine, delivers Falcon 9's payload to the desired orbit.



Landing Pad in Ocean.

A New Trick May Help SpaceX Launch Bigger Payloads, Save More Rockets (businessinsider.com)